

Dhwani Auvnish Chande

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Education

University of Illinois Chicago

Master of Science in Computer Science

Chicago, IL

Aug. 2025 – May 2027

GPA: 3.40/4.00

Relevant Graduate Coursework: Language Processing, Big Data Mining, Advanced Computer Vision, Computer Graphics I, Introduction to Data Science

Upcoming Fall 2026: Visual Data Science, Vector Databases

SVKM's Dwarkadas J. Sanghvi College of Engineering

Bachelor of Technology in Artificial Intelligence and Machine Learning; Honors in Immersive Technologies Aug. 2021 – May 2025

Mumbai, India

Relevant Undergraduate Coursework: Advanced Data Structures and Algorithms, Database Management Systems, Operating Systems, Deep Learning, Computer Vision, Natural Language Processing, Machine Learning Operations, Large Language Models, High Performance Computing, Reinforcement Learning, Ethical AI, Image Generative AI

Technical Skills

Languages: Python, Java, SQL, C/C++, JavaScript

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, TensorFlow, NumPy, Pandas

Computer Vision and 3D: OpenCV, COLMAP, 3D Gaussian Splatting, Mip-Splatting, Novel View Synthesis

Backend and Testing: FastAPI, REST APIs, Pydantic, pytest, Docker

Data Mining, Engineering, and Visualization: PySpark, MLlib, mlxtend, GeoPandas, D3.js, Matplotlib

Developer Tools and Platforms: Git, GitHub, GitHub Actions, Linux/Bash, Google Colab, Weights & Biases, Hugging Face Spaces, ffmpeg

Projects

Natural Language to Bash Translation using LLMs | [GitHub](#) | [Model](#) | [Demo](#)

Aug. 2025 – Dec. 2025

Technologies: Python, PyTorch, Hugging Face Transformers, FastAPI, Pydantic, Docker, pytest, GitHub Actions

- Fine-tuned Llama-3.2-1B-Instruct and Qwen2.5-Coder-0.5B-Instruct on 40,639 natural-language-to-Bash command pairs using full-parameter supervised fine-tuning, and published the Qwen model on Hugging Face.
- Evaluated both models on a 300-example set using exact-match and embedding-based semantic metrics; Qwen achieved 13.67% exact match, 60.33% semantic match, and 0.776 average cosine similarity.
- Compared eight functional-equivalence heuristics across 600 labeled command pairs; execution-assisted embedding evaluation achieved 90% accuracy and 0.89 F1, outperforming lexical and standalone LLM-based methods.
- Built and containerized a FastAPI inference service with Pydantic request validation, health monitoring, latency logging, and single- and batch-translation endpoints supporting up to 10 instructions per request.
- Developed a 20-test pytest suite covering endpoint behavior, response schemas, input-validation boundaries, and batch limits; configured GitHub Actions CI to run the suite on pushes and pull requests to the main branch.

3D Gaussian Splatting: Novel View Synthesis and Scale-Aware Rendering | [GitHub](#)

Jan. 2026 – May 2026

Technologies: Python, PyTorch, 3D Gaussian Splatting, Mip-Splatting, COLMAP, OpenCV, ffmpeg

- Built an end-to-end experimentation pipeline for 3D Gaussian Splatting and Mip-Splatting, automating COLMAP preprocessing, model training, multi-resolution rendering, quantitative evaluation, and qualitative visualization.
- On the benchmark scene, Mip-Splatting improved PSNR from 21.13 dB to 28.09 dB and SSIM from 0.702 to 0.896 while reducing LPIPS from 0.244 to 0.088, a 64% reduction in perceptual error.
- Reconstructed a self-captured scene from iPhone video by extracting frames with ffmpeg, estimating camera poses and sparse geometry with COLMAP, and training both rendering methods on the processed scene.
- Developed crop-level comparisons and absolute-difference heatmaps to identify larger rendering errors around foliage, object boundaries, and reflective surfaces.

Geospatial Analysis of Urban Food Deserts | [GitHub](#) | [Dashboard](#)

Aug. 2025 – Dec. 2025

Technologies: Python, pandas, NumPy, scikit-learn, imbalanced-learn, GeoPandas, Census API, D3.js, HTML, CSS, JavaScript

- Implemented affordability-modeling and digital-access analyses for a five-member study of 1,314 Cook County census tracts, integrating USDA food-access records with ACS broadband, demographic, income, SNAP, and transportation data.
- Engineered 12 affordability and demographic features and benchmarked Logistic Regression, Random Forest, Gradient Boosting, SVM, and Naive Bayes classifiers; balanced Logistic Regression achieved 0.804 ROC-AUC and 66.7% food-desert recall.
- Constructed a Digital Barrier Index from standardized broadband adoption, senior-population, SNAP-participation, and vehicle-access measures to examine the relationship between digital exclusion and food-desert status.

- Applied SMOTE only to the training split to address a 3.9% positive-class prevalence, increasing food-desert recall from 0% to 73% while achieving 0.825 ROC-AUC on the original test set.
- Evaluated model stability using stratified five-fold cross-validation, obtaining a mean ROC-AUC of 0.838 with a standard deviation of 0.026, and validated group differences using point-biserial correlation and a Mann–Whitney U test.
- Built and deployed an interactive D3.js dashboard on GitHub Pages, implementing census-tract maps, filtering controls, cluster visualizations, and demographic trend charts for exploratory analysis by non-technical users.

FairLend Miners: Fairness-Aware Mortgage Lending Analysis | [GitHub](#) | [Paper](#)

Jan. 2026 – May 2026

Technologies: Python, pandas, NumPy, mlxtend, FP-Growth, Association Rule Mining, Parquet, Matplotlib, Seaborn

- Implemented the FP-Growth stage of a three-member mortgage-lending fairness audit on 103,481 HMDA 2023 applications, transforming 11 categorical loan, financial, and demographic attributes into transactional records for association-rule mining.
- Configured mlxtend FP-Growth with 10% minimum support and a maximum itemset length of four, then generated association rules requiring at least 50% confidence and lift of 1.0 or greater.
- Compared standard equal-frequency and fairness-aware binnings, mining 819 versus 854 frequent itemsets and 2,214 versus 2,454 association rules, respectively.
- Identified high debt-to-income ratio as the strongest high-support denial predictor, with 10.7% support, 67.2% confidence, and 2.81 lift.
- Found no explicit demographic denial rules above the 10% support threshold; the team subsequently applied K-Means clustering and a cluster-level disparate-impact audit to investigate disparities not captured by global association rules.
- Built comparative visualizations and exported complete and denial-specific rule sets to CSV to support reproducible analysis and the accompanying research paper.

Publications

A. Rathod, **D. Chande**, and H. Nagda, “Auditing Discriminatory Patterns in Mortgage Lending Through Association Rules and Fair Binning,” *arXiv preprint arXiv:2606.12435*, 2026. [Paper](#)

S. Shah, **D. Chande**, K. Pandya, K. Gopani, A. Gawade, N. Rathod, and A. Florence, “Revolutionizing Culinary Experiences: AI-Driven Ingredient Recognition and Personalized Recipe Recommendation System,” *Library Progress International*, vol. 44, no. 3, 2024. [DOI](#)

Y. Shah, V. Parekh, **D. Chande**, S. Khot, A. Gawade, N. Rathod, and A. Florence, “Smart Wardrobe: A Comprehensive Approach to Personalized Clothing Recommendation with the Use of Nearest Neighbor Model,” *Journal of Electrical Systems*, vol. 20, no. 10s, pp. 6898–6907, 2024. [Publication](#)

Certifications

Fundamentals of Accelerated Computing with CUDA C/C++, NVIDIA, Sep. 2024

Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Jun. 2024

Unsupervised Learning, Recommenders, Reinforcement Learning, DeepLearning.AI and Stanford University, Jun. 2024